

Qualification Pack



Instrumentation Technician (Mineral Sands)

QP Code: MIN/Q3302

Version: 1.0

NSQF Level: 4.5

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MIN/Q3302: Instrumentation Technician (Mineral Sands)

Brief Job Description

An Instrumentation Technician (Mineral Sands) is responsible for monitoring, maintaining and repairing the processing tools and equipment used in rare earth mining.

Personal Attributes

The individual should be physically fit to work for long durations with concentration. The person should have attention to detail, problem-solving and organizational skills. The individual should also have appropriate verbal and written communication skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [MIN/N3303: Perform inspection, testing, calibration and troubleshooting of measuring and control equipment](#)
2. [MIN/N3304: Carry out preventive, breakdown, scheduled and predictive maintenance of mining equipment](#)
3. [MIN/N1810: Follow the applicable health, safety and environmental guidelines \(Placer Mining\)](#)
4. [MIN/N1811: Work collaboratively and effectively in a team](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Mining
Sub-Sector	Engineering Services
Occupation	Instrumentation and Control Systems
Country	India
NSQF Level	4.5
Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7311.0101

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Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) (in Electrical /Instrumentation) OR Completed 3 year diploma after 10th (in Electrical /Instrumentation) OR 10th grade pass (with 3 year of any combination of NTC/NAC/CITS or equivalent (in Electrical /Instrumentation))
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	14/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-4.5-MG-02176-2024-V1-SCMS
NQR Version	1.0

Remarks:

480 Hours Duration

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MIN/N3303: Perform inspection, testing, calibration and troubleshooting of measuring and control equipment

Description

This occupational standards unit is about the inspection, testing, calibration and troubleshooting of measuring and control equipment

Scope

The scope covers the following :

- Prepare and use the testing tools for fault elimination
- Test the measuring tools
- Analyze the process equipment/ system
- Calibrate the measuring tools

Elements and Performance Criteria

Prepare and use the testing tools for fault elimination

To be competent, the user/individual on the job must be able to:

- PC1.** select the appropriate testing tool for use, such as multimeter, clamp meter, Piezo meter, sensor etc., to measure parameters like Voltage, Current, Resistance, Pressure, Flow, Level, Depth, Temperature etc
- PC2.** check the testing tools for faults, wear and tear or damage before use
- PC3.** use the testing equipment as per the Original Equipment Manufacturer (OEM) manual/ instructions
- PC4.** validate the calibration and traceability of testing tools against the applicable national standards
- PC5.** validate the calibration and traceability of testing tools against the applicable national standards
- PC6.** validate the performance of electrical and electronics devices using the testing tools in the circuit/functional block diagram (FBD) - Centrifuge, Filter Press, Crystallizer, Auto Sampler, Mass Flow Meter, Magnetic Flow Meter, Water Flow Meter, Air Flow Meter, Oil Flow meter, Density meter, Load cells etc
- PC7.** diagnose faults to assess the operation of instrumentation system
- PC8.** undertake appropriate corrective measures to eliminate the fault to restore the normal performance of the system

Test the measuring tools

To be competent, the user/individual on the job must be able to:

- PC9.** check the measuring tools, such as flow meter, level indicator, pH meter, density meter, pressure gauge/ temperature gauge, pinch control valve, butterfly valve etc., for faults, wear and tear or damage
- PC10.** check the zero, span and range on indicators/controllers using the recommended configuration/ OEM manual

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- PC11.** validate the actual parameter with the set value as the per requirement of the respective processes
- PC12.** calibrate the tools following the standard procedure/ OEM manual in case of any deviations
- PC13.** validate the operation and function of tools after calibrations
- PC14.** check the performance and reliability of the tools with the environmental conditions i.e. extreme temperature, humidity, etc., before and after testing
- PC15.** isolate the faulty measuring tools from the complete system following the appropriate safety measures
- PC16.** use the relevant testing devices to diagnose faults of operational instruments and systems i.e. sensors, transmitters, controllers, power circuits, power supplies, indicators, analyzers etc
- PC17.** replace the faulty meter/tool with the new one
- PC18.** validate and calibrate the operational and functional maintenance or service of the tool

Analyze the process equipment/ system

To be competent, the user/individual on the job must be able to:

- PC19.** identify the faults and verify the results against the operational parameters, i.e. pressure, temperature, flow, level, conductivity, weight, mass, volume, density, pH Value, etc
- PC20.** identify the fault with respect to its sequence of operation and interlock with different process parameters as per its design and operation
- PC21.** detect the existing and potential faults and report the same to concerned authority following the standard procedure
- PC22.** repair or replace the faulty component, as appropriate
- PC23.** check the functionality of the system after eliminating fault as per the operations and functions of the process
- PC24.** maintain the maintenance log book and history card

Calibrate the measuring tools

To be competent, the user/individual on the job must be able to:

- PC25.** calibrate the tools for Mass flow, Volume flow, temperature, pressure, level, pH etc., as per the standards procedure/ OEM manual
- PC26.** follow the master calibration standard, serviceability and traceability at regular intervals with the applicable national standard
- PC27.** calibrate the measuring tools and ensure they are free from damage
- PC28.** identify and resolve the error of the tools being calibrated to increase their accuracy and efficiency
- PC29.** ensure compliance with the applicable quality control procedures
- PC30.** monitor the function of the system and update the supervisor about the progress or any delay in resolving the problem
- PC31.** recommission the equipment according to the SOP
- PC32.** maintain the relevant documents, e.g. the action plan with the relevant information

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** the uses and benefits of industrial sensors and transducers
- KU2.** cable end termination, glanding, ferruling and tagging
- KU3.** the standard procedure for the installation of measuring instruments
- KU4.** how to maintain log book/ calibration/ maintenance records
- KU5.** the use of relevant measuring tools
- KU6.** the appropriate parameters for measuring equipment and components for their optimum functioning
- KU7.** the appropriate techniques for checking the calibration of measuring tools
- KU8.** the techniques to check the faults in the measuring tools
- KU9.** the troubleshooting of common errors/faults found in the measuring tools and functional operating blocks
- KU10.** the importance of error-free performance/ accuracy of measuring tools
- KU11.** the hazards related to the calibration of the measuring tools
- KU12.** the applicable instrumentation principles such as controlling density, level, flow, temperature, pH, Sensor and Transducer, and arrangement of materials
- KU13.** the principles of hydraulic/pneumatic flow and instrumentation
- KU14.** the principles and application for assessing the process of instrument, equipment/components
- KU15.** the instrumentation system
- KU16.** the relevant equipment for the inspection and testing of the instrumentation system
- KU17.** the procedure for repairing faults in the instrumentation system
- KU18.** the techniques for dismantling, testing and re-assembling the instrumentation system
- KU19.** the procedure of checking and verifying the operational function of FOB
- KU20.** the appropriate documentation requirements, such as job card, progress report, incident report, calibration and maintenance log book, calibration Labels, test reports, nonconforming calibration reports, calibration certificates, etc
- KU21.** the importance and process of maintaining the service records e.g. test and calibration results
- KU22.** how to resolve variations between test results and operational parameters
- KU23.** the importance and process of reporting the unresolved faults
- KU24.** the process of carrying out zero-and-span checks on tools
- KU25.** the use of instrumentation control equipment, such as mechanical system, load cells/strain gauges, transducers, Water/Air/Oil flow meters, etc
- KU26.** how to check the speed measurement and control equipment, such as valves and valve mechanisms including control valves, valve actuators and positioners, Butterfly Valves, Pinch Control Valves, Isolation Valves, Gauges, pressure transmitter, PID controllers etc
- KU27.** the use of various testing and calibration tools, such as pressure gauge, standard test gauge, micrometer, insulation testers, calibrated weights, Vernier calliper, dead weight tester, manometer, gyroscope, etc
- KU28.** the equipment specifications and their use
- KU29.** the use of calibration to identify the amount of drift to analyze instrument stability and measurement consistency

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- KU30.** the importance of using tools, apparatus and equipment approved by Directorate General Of Mines Safety (DGMS)/competent authority
- KU31.** the importance of regular cleaning and maintenance of tools and equipment
- KU32.** how to isolate the power tools and equipment during emergencies
- KU33.** the importance and use of safe tripping /interlocking mechanism to prevent electrical fires
- KU34.** the importance of providing double earthing 3-phase equipment
- KU35.** the importance of ensuring that installation of cables and equipment as per IER-1910, IEA1956, and NEC1985 & ESA1948

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** listen attentively to understand the information/instructions being shared
- GS3.** read the appropriate literature to learn about the latest technology related to the field of work
- GS4.** communication politely and professionally
- GS5.** use effective questioning techniques to extract maximum information about different tasks
- GS6.** plan daily tasks appropriately to ensure their timely completion
- GS7.** undertake different activities as per the applicable task lists and schedules with minimum supervision
- GS8.** make appropriate decisions concerning the area of work
- GS9.** assist in identifying and implementing appropriate solutions to work-related problems
- GS10.** follow the organizational protocol during emergencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare and use the testing tools for fault elimination</i>	8	15	-	6
PC1. select the appropriate testing tool for use, such as multimeter, clamp meter, Piezo meter, sensor etc., to measure parameters like Voltage, Current, Resistance, Pressure, Flow, Level, Depth, Temperature etc	1	3	-	1
PC2. check the testing tools for faults, wear and tear or damage before use	1	3	-	1
PC3. use the testing equipment as per the Original Equipment Manufacturer (OEM) manual/ instructions	1	1	-	1
PC4. validate the calibration and traceability of testing tools against the applicable national standards	1	1	-	1
PC5. validate the calibration and traceability of testing tools against the applicable national standards	1	3	-	1
PC6. validate the performance of electrical and electronics devices using the testing tools in the circuit/functional block diagram (FBD) - Centrifuge, Filter Press, Crystallizer, Auto Sampler, Mass Flow Meter, Magnetic Flow Meter, Water Flow Meter, Air Flow Meter, Oil Flow meter, Density meter, Load cells etc	1	1	-	-
PC7. diagnose faults to assess the operation of instrumentation system	1	1	-	-
PC8. undertake appropriate corrective measures to eliminate the fault to restore the normal performance of the system	1	2	-	1
<i>Test the measuring tools</i>	8	16	-	7
PC9. check the measuring tools, such as flow meter. level indicator, pH meter, density meter, pressure gauge/ temperature gauge, pinch control valve, butterfly valve etc., for faults, wear and tear or damage	1	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. check the zero, span and range on indicators/controllers using the recommended configuration/ OEM manual	1	3	-	1
PC11. validate the actual parameter with the set value as the per requirement of the respective processes	-	1	-	-
PC12. calibrate the tools following the standard procedure/ OEM manual in case of any deviations	1	1	-	1
PC13. validate the operation and function of tools after calibrations	-	1	-	-
PC14. check the performance and reliability of the tools with the environmental conditions i.e. extreme temperature, humidity, etc., before and after testing	1	3	-	1
PC15. isolate the faulty measuring tools from the complete system following the appropriate safety measures	1	1	-	1
PC16. use the relevant testing devices to diagnose faults of operational instruments and systems i.e. sensors, transmitters, controllers, power circuits, power supplies, indicators, analyzers etc	1	1	-	1
PC17. replace the faulty meter/tool with the new one	1	1	-	1
PC18. validate and calibrate the operational and functional maintenance or service of the tool	1	1	-	-
Analyze the process equipment/ system	6	10	-	3
PC19. identify the faults and verify the results against the operational parameters, i.e. pressure, temperature, flow, level, conductivity, weight, mass, volume, density, pH Value, etc	1	2	-	1
PC20. identify the fault with respect to its sequence of operation and interlock with different process parameters as per its design and operation	1	2	-	1
PC21. detect the existing and potential faults and report the same to concerned authority following the standard procedure	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. repair or replace the faulty component, as appropriate	1	1	-	-
PC23. check the functionality of the system after eliminating fault as per the operations and functions of the process	1	3	-	1
PC24. maintain the maintenance log book and history card	1	1	-	-
<i>Calibrate the measuring tools</i>	8	9	-	4
PC25. calibrate the tools for Mass flow, Volume flow, temperature, pressure, level, pH etc., as per the standards procedure/ OEM manual	1	2	-	1
PC26. follow the master calibration standard, serviceability and traceability at regular intervals with the applicable national standard	1	1	-	1
PC27. calibrate the measuring tools and ensure they are free from damage	1	1	-	1
PC28. identify and resolve the error of the tools being calibrated to increase their accuracy and efficiency	1	1	-	1
PC29. ensure compliance with the applicable quality control procedures	1	1	-	-
PC30. monitor the function of the system and update the supervisor about the progress or any delay in resolving the problem	1	1	-	-
PC31. recommission the equipment according to the SOP	1	1	-	-
PC32. maintain the relevant documents, e.g. the action plan with the relevant information	1	1	-	-
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3303
NOS Name	Perform inspection, testing, calibration and troubleshooting of measuring and control equipment
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Instrumentation and Control Systems
NSQF Level	4.5
Credits	7
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N3304: Carry out preventive, breakdown, scheduled and predictive maintenance of mining equipment

Description

This occupational standards unit is about carrying out the preventive, breakdown, scheduled and predictive maintenance of rare earth mining equipment

Scope

The scope covers the following :

- Conduct preventive maintenance
- Conduct breakdown maintenance
- Conduct condition monitoring
- Conduct scheduled maintenance
- Conduct predictive maintenance

Elements and Performance Criteria

Conduct preventive maintenance

To be competent, the user/individual on the job must be able to:

- PC1.** schedule and perform regular preventive maintenance of instrumentation systems/ equipment/ components, e.g. connection to terminal blocks, cable lugs, relays, Modules, Switches and Conductor, etc.
- PC2.** carry out cleaning to ensure a dust and moisture free environment for the instrumentation system
- PC3.** lubricate the moving machines parts of reduce wear and tear
- PC4.** adjust the process parameters for optimal equipment effectiveness
- PC5.** repair or replace the defective parts identified during preventive maintenance
- PC6.** carry out appropriate documentation concerning the preventive maintenance activities

Conduct breakdown maintenance

To be competent, the user/individual on the job must be able to:

- PC7.** perform breakdown for the equipment parts that experience breakdown, following the maintenance plan
- PC8.** use the relevant testing tools to measure and control pressure, temperature, flow, level, motion, force, etc.
- PC9.** use the appropriate tools to diagnose faults in the circuit
- PC10.** coordinate with the appropriate person concerning the breakdown maintenance emergencies
- PC11.** adjust the system components and replace the defective parts
- PC12.** troubleshoot for test failure to determine root causes and develop process plan to avoid faults
- PC13.** carry out appropriate documentation concerning the breakdown maintenance activities

Conduct condition monitoring

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To be competent, the user/individual on the job must be able to:

- PC14.** use the appropriate critical equipment for condition monitoring, e.g. mass flow meter, weighbridge, FBD, Centrifuge, Press Filter, etc
- PC15.** assess the equipment status and operation
- PC16.** conduct condition monitoring for the relevant equipment following the OEM manual/ in-house standard procedure
- PC17.** collect and analyze the condition monitoring data for further actions and scheduling
- PC18.** perform maintenance as per condition monitoring
- PC19.** carry out appropriate documentation concerning the maintenance activities

Conduct scheduled maintenance

To be competent, the user/individual on the job must be able to:

- PC20.** identify and prioritize the scheduled maintenance of the relevant instrumentation systems/equipment/ components
- PC21.** examine the faulty parts scheduled for replacement and take appropriate corrective action
- PC22.** undertake appropriate adjustments using the calibration tools
- PC23.** document the prescribed procedures and operational specifications
- PC24.** recommission the equipment following the standard procedure

Conduct predictive maintenance

To be competent, the user/individual on the job must be able to:

- PC25.** monitor the critical process parameters through the data acquired from field sensors
- PC26.** analyze the key trends concerning the faults and update the deviations
- PC27.** coordinate with the concerned authority or expert to avoid the deviation and optimize maintenance strategy
- PC28.** perform maintenance activities as per predictive maintenance fault identification

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of preventive maintenance to minimize breakdowns and maximize the equipment efficiency
- KU2.** the use of appropriate documents, e.g. job card, work permit, risk assessment, sign-on/off, maintenance logs, etc.
- KU3.** the health and safety precautions for the maintenance process
- KU4.** how to minimize risks from the hazards during the equipment maintenance
- KU5.** the use of appropriate Personal Protective Equipment (PPE) for the maintenance activities
- KU6.** the importance of following the Original Equipment Manufacturer (OEM) manual for equipment maintenance
- KU7.** the appropriate methods to identify faults in the equipment
- KU8.** the evaluation of different types of information for fault diagnoses, such as operator-provided information, equipment self-diagnosis, and testing tool measurements
- KU9.** the use of various testing tools, such as multimeter, clamp meter, flow meter etc

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- KU10.** the use of appropriate aids for fault diagnosis, such as logic diagrams, Pareto analysis, manufacturers' manuals, probability charts, computer-aided test equipment, electronic aids, etc
- KU11.** the application of various fault finding techniques, such as half-split technique, input/output technique, injection and sampling, six-point technique, emergent sequence, unit substitution, function/performance testing, equipment self-diagnostics, etc
- KU12.** how to evaluate sensory conditions, i.e. flow, temperature, pressure, density
- KU13.** how to evaluate the risk of running the equipment with the displayed fault, and the effects the fault may have on health and safety
- KU14.** the handling and maintenance of different testing tools
- KU15.** the appropriate precautions to be taken to prevent Electrostatic Discharge (ESD) damage to electronic circuits and components
- KU16.** the basic principles of operation of testing tools
- KU17.** the importance of individual units/ components such as gauges and meters and how they interact
- KU18.** the importance of ensuring Lock Out and Tag Out (LOTO)
- KU19.** the process of identifying and selecting instrument sensors, including the identification of their markings, calibration information, component values, operating parameters and working range
- KU20.** the use of appropriate fitting tools to avoid faulty readings, poor flow past sensor, blockages, incorrect wiring, poor insulation or incorrect materials
- KU21.** the process of installing and connecting external wiring and components safely to avoid electronic interference or mechanical damage
- KU22.** how to carry out visual checks of the tools, including checks for leaks, security of joints and damage
- KU23.** the relevant replacement parts, materials and other items required for the maintenance process
- KU24.** the techniques used to dismantle/ assemble integrated equipment i.e. release of pressure/force, and clamped and crimped connections
- KU25.** the importance and process of attaching identification marks/labels to remove components or cables
- KU26.** how to adopt the Plan-Do-Check-Act-Implement (PDCAI) approach for maintenance
- KU27.** the applicable Standard Operating Procedures (SOPs), maintenance schedules and plans
- KU28.** the benefits and methods of resource optimization

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to get the latest updates about the field of work
- GS3.** listen attentively to understand the information/ instructions being shared
- GS4.** communicate politely and professionally
- GS5.** plan and undertake various tasks to ensure their timely completion



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- GS6.** evaluate all possible solutions to a problem and select the best one
- GS7.** use effective questioning techniques to extract maximum information about different tasks
- GS8.** undertake different activities as per the applicable task lists and schedules with minimum supervision
- GS9.** make appropriate decisions concerning the area of work
- GS10.** follow the organizational protocol during emergencies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conduct preventive maintenance</i>	7	13	-	5
PC1. schedule and perform regular preventive maintenance of instrumentation systems/ equipment/ components, e.g. connection to terminal blocks, cable lugs, relays, Modules, Switches and Conductor, etc.	1	1	-	1
PC2. carry out cleaning to ensure a dust and moisture free environment for the instrumentation system	2	3	-	1
PC3. lubricate the moving machines parts of reduce wear and tear	1	2	-	-
PC4. adjust the process parameters for optimal equipment effectiveness	1	2	-	1
PC5. repair or replace the defective parts identified during preventive maintenance	1	2	-	1
PC6. carry out appropriate documentation concerning the preventive maintenance activities	1	3	-	1
<i>Conduct breakdown maintenance</i>	7	12	-	5
PC7. perform breakdown for the equipment parts that experience breakdown, following the maintenance plan	1	1	-	1
PC8. use the relevant testing tools to measure and control pressure, temperature, flow, level, motion, force, etc.	1	2	-	1
PC9. use the appropriate tools to diagnose faults in the circuit	1	2	-	-
PC10. coordinate with the appropriate person concerning the breakdown maintenance emergencies	1	2	-	1
PC11. adjust the system components and replace the defective parts	1	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. troubleshoot for test failure to determine root causes and develop process plan to avoid faults	1	1	-	1
PC13. carry out appropriate documentation concerning the breakdown maintenance activities	1	3	-	1
<i>Conduct condition monitoring</i>	6	10	-	4
PC14. use the appropriate critical equipment for condition monitoring, e.g. mass flow meter, weighbridge, FBD, Centrifuge, Press Filter, etc	1	2	-	1
PC15. assess the equipment status and operation	1	1	-	-
PC16. conduct condition monitoring for the relevant equipment following the OEM manual/ in-house standard procedure	1	2	-	1
PC17. collect and analyze the condition monitoring data for further actions and scheduling	1	1	-	1
PC18. perform maintenance as per condition monitoring	1	1	-	1
PC19. carry out appropriate documentation concerning the maintenance activities	1	3	-	-
<i>Conduct scheduled maintenance</i>	6	8	-	3
PC20. identify and prioritize the scheduled maintenance of the relevant instrumentation systems/equipment/ components	2	3	-	1
PC21. examine the faulty parts scheduled for replacement and take appropriate corrective action	1	2	-	1
PC22. undertake appropriate adjustments using the calibration tools	1	1	-	1
PC23. document the prescribed procedures and operational specifications	1	1	-	-
PC24. recommission the equipment following the standard procedure	1	1	-	-
<i>Conduct predictive maintenance</i>	4	7	-	3



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC25. monitor the critical process parameters through the data acquired from field sensors	1	2	-	1
PC26. analyze the key trends concerning the faults and update the deviations	1	1	-	1
PC27. coordinate with the concerned authority or expert to avoid the deviation and optimize maintenance strategy	1	2	-	-
PC28. perform maintenance activities as per predictive maintenance fault identification	1	2	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	MIN/N3304
NOS Name	Carry out preventive, breakdown, scheduled and predictive maintenance of mining equipment
Sector	Mining
Sub-Sector	Engineering Services
Occupation	Instrumentation and Control Systems
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

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MIN/N1810: Follow the applicable health, safety and environmental guidelines (Placer Mining)

Description

This unit is about following the applicable health, safety and environmental guidelines.

Scope

The scope covers the following :

- Maintain personal health and safety
- Assist in hazard management
- Work on equipment and machinery
- Check the first aid box, firefighting and safety equipment
- Report incidents

Elements and Performance Criteria

Maintain personal health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers
- PC2.** ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments
- PC3.** clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes
- PC4.** follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)
- PC5.** follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking
- PC6.** use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose
- PC7.** identify six directional hazards at the workplace and take decisions accordingly
- PC8.** follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas
- PC9.** undertake the Take-5 (Personal Risk Assessment) before the commencement of any work
- PC10.** lift objects carefully maintaining the correct body posture to avoid injury
- PC11.** use self-rescue apparatus appropriately when required
- PC12.** work in the assigned work zone in the designated work area
- PC13.** follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc
- PC14.** wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant
- PC15.** use caution signs as applicable and obey safety signs, stickers, and tags

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PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact

PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help

Assist in hazard management

To be competent, the user/individual on the job must be able to:

PC18. label and store chemicals appropriately

PC19. display the MSDS for the hazardous chemicals in storage and handling areas

PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines

PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre

PC22. maintain regular and open communication with co-workers and the employer

Work on equipment and machinery

To be competent, the user/individual on the job must be able to:

PC23. check and ensure every instrument, apparatus and equipment is AERB approved

PC24. operate the appropriate tools, equipment and machinery with proper training and authorization

Check the first aid box, firefighting, and safety equipment

To be competent, the user/individual on the job must be able to:

PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible

PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas

PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors

PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite

Report incidents

To be competent, the user/individual on the job must be able to:

PC29. identify and report potential hazards/ unsafe conditions and practices

PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.

PC31. report the emergencies and any safety incidents as per the organizational procedure

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. different types of chemical plants and their operations

KU2. the importance of time-keeping, discipline and punctuality

KU3. various chemical processes carried out in a rare earth extraction plant

KU4. the impact of violating applicable safety procedures

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- KU5.** the relevant quality norms
- KU6.** different types of individual rare earth chemical processes and the use of associated equipment
- KU7.** different types of effluent treatment processes and the use of associated equipment
- KU8.** the use of different water purifying and waste treatment agents
- KU9.** the hazards and the appropriate safety measures in acid/ alkali handling activities and usage of relevant PPE
- KU10.** hazards and safety aspects involved in acid/ alkali handling activities and usage of relevant PPE
- KU11.** the internal safety organization and the role of the safety committee, workforce inspector and AERB, etc
- KU12.** the importance of sensitization towards different genders and persons with disabilities (PWD)
- KU13.** the Importance of following infection control policies, '5-S' practices, and waste management as prescribed by regulatory authorities like the Atomic Energy Regulatory Board (AERB), United Nations Development Group (UNDG), etc.
- KU14.** the importance of water/material/energy conservation and management
- KU15.** the importance of Social Management Plan (SMP) and Environment Management Plan (EMP) prepared by the organization, as per DGMS guideline
- KU16.** the importance of wearing own TLD or pocket dosimeter (to be obtained from the Health Physics Unit) in the active area
- KU17.** the importance of staying alert and attentive to the work surroundings
- KU18.** the importance of staying in the assigned work zone in the designated work area
- KU19.** the benefits of making the safety rules part of new employee orientation
- KU20.** the importance of contacting the supervisor or manager for instruction, guidance, or training when required

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal health and safety</i>	16	24	-	11
PC1. follow the appropriate hand hygiene measures, i.e., washing hands with EDTA/Soda Ash (Ethylene Diamine Tetra Acetate/Sodium Carbonate), soap and water or alcohol-based hand sanitizers	1	2	-	1
PC2. ensure the plants are provided with a dilution ventilation system to avoid the undue accumulation of chemical vapours in working environments	1	1	-	1
PC3. clean and disinfect the regularly used objects, such as mobile phones, wireless communication systems keys, wallets and work passes	1	1	-	-
PC4. follow the Safety Management Plan (SMP) and Emergency Management Plan (EMP)	1	2	-	-
PC5. follow sanitation measures in the work area, e.g., avoiding eating, drinking, chewing and smoking	1	2	-	-
PC6. use the appropriate PPE, e.g. using an appropriate mask to cover the mouth and nose	1	2	-	-
PC7. identify six directional hazards at the workplace and take decisions accordingly	1	2	-	1
PC8. follow appropriate safety measures to avoid exposure to radiation, e.g. limiting stay in the radiation area and avoiding working with open wounds or exposed body parts in radiation areas	1	2	-	1
PC9. undertake the Take-5 (Personal Risk Assessment) before the commencement of any work	1	2	-	1
PC10. lift objects carefully maintaining the correct body posture to avoid injury	-	1	-	-
PC11. use self-rescue apparatus appropriately when required	1	1	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. work in the assigned work zone in the designated work area	1	1	-	1
PC13. follow the recommended safety measures while working at haul roads, heights, dumps, sump areas, stockyards, near moving parts, etc	1	1	-	1
PC14. wear Thermo luminescent Dosimeter (TLD) Badge for personal monitoring in rare earth extraction plant	1	1	-	1
PC15. use caution signs as applicable and obey safety signs, stickers, and tags	1	1	-	-
PC16. identify the environmental impact of operations related to mining, mineral separation plant and rare earth extraction plant and take steps to reduce the impact	1	1	-	1
PC17. check for contamination of sensitive body parts, e.g. face, eyes etc., and seek immediate medical help	1	1	-	1
<i>Assist in hazard management</i>	5	8	-	4
PC18. label and store chemicals appropriately	1	1	-	1
PC19. display the MSDS for the hazardous chemicals in storage and handling areas	1	2	-	1
PC20. educate the personnel at the workplace about the safety requirements and follow the applicable safety guidelines	1	1	-	-
PC21. check the body and clothing for contamination at the end of radioactive work, and decontaminate and take a shower at the decontamination centre	1	2	-	1
PC22. maintain regular and open communication with co-workers and the employer	1	2	-	1
<i>Work on equipment and machinery</i>	2	4	-	1
PC23. check and ensure every instrument, apparatus and equipment is AERB approved	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. operate the appropriate tools, equipment and machinery with proper training and authorization	1	2	-	1
<i>Check the first aid box, firefighting, and safety equipment</i>	4	10	-	2
PC25. check the availability of first aid kits and fire extinguishers, and ensure they are easily accessible	1	3	-	-
PC26. ensure the availability of emergency showers and eye washers in the chemical handling, unloading and pumping areas	1	2	-	-
PC27. monitor the work environment at appropriate intervals for hydrogen chloride and hydrocarbons using portable monitors	1	2	-	1
PC28. operate the appropriate type of fire extinguishers to control different types of fire at the worksite	1	3	-	1
<i>Report incidents</i>	3	4	-	2
PC29. identify and report potential hazards/ unsafe conditions and practices	1	2	-	-
PC30. follow the appropriate emergency procedure during emergencies, such as power failures, cyclones, etc.	1	1	-	1
PC31. report the emergencies and any safety incidents as per the organizational procedure	1	1	-	1
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1810
NOS Name	Follow the applicable health, safety and environmental guidelines (Placer Mining)
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

MIN/N1811: Work collaboratively and effectively in a team

Description

This unit is about the importance of teamwork in the workplace.

Scope

The scope covers the following :

- Work effectively with co-workers
- Communicate effectively with co-workers
- Practice inclusiveness at work

Elements and Performance Criteria

Work effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC1.** identify own direct and indirect responsibilities, e.g. duties concerning team members
- PC2.** undertake appropriate activities to promote cooperation and good relationships with co-workers
- PC3.** mentor and assist subordinates in the execution of their work responsibilities
- PC4.** identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures
- PC5.** follow the recommended practices to avoid and resolve conflicts at work
- PC6.** interact effectively and cooperate with co-workers
- PC7.** manage workplace conflicts, ensuring no impact on work
- PC8.** identify and take responsibility for own mistakes
- PC9.** follow the organizational policies and procedures
- PC10.** follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.
- PC11.** exchange feedback with peers and other team members to improve the quality of work

Communicate effectively with co-workers

To be competent, the user/individual on the job must be able to:

- PC12.** follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel
- PC13.** communicate clearly and politely to ensure effective communication with co-workers
- PC14.** practice active listening to understand the conveyed information and instructions
- PC15.** practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication
- PC16.** maintain the professional standard of communication, e.g. avoiding the use of offensive language at work

Practice inclusiveness at work

To be competent, the user/individual on the job must be able to:

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- PC17.** empathize with Persons with Disabilities (PwD)
- PC18.** practice gender equality at work
- PC19.** maintain respectful behaviour and etiquette at work
- PC20.** listen to and acknowledge the feelings, concerns, opinions, and ideas of others

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the importance of teamwork and the consequence of poor teamwork on project outcomes, timelines, safety, etc
- KU2.** the duties and rights of workers
- KU3.** the importance of creating a healthy and cooperative work environment
- KU4.** the importance of proper and effective communication and the adverse effects of poor communication on quality, timeliness, safety, etc
- KU5.** different modes of communication used at work
- KU6.** the importance and need of supporting co-workers facing problems for the smooth functioning of work
- KU7.** the fundamental concept of gender equality
- KU8.** how to recognize and be sensitive to issues of disability, culture and gender
- KU9.** how to show empathy to PwD appropriately
- KU10.** legislation, policies, and procedures relating to gender sensitivity and cultural diversity including their impact on the area of operation
- KU11.** the appropriate behaviour to be practised with coworkers
- KU12.** the appropriate communication skills and leadership qualities
- KU13.** how to be open to listening to and understanding the other person's point of view

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete appropriate documentation concerning work
- GS2.** make appropriate decisions concerning the area of work with effective reasoning
- GS3.** read and explain manuals, health, and safety instructions, memos, etc
- GS4.** plan and organize work effectively
- GS5.** detect problems at work and use reasoning skills to resolve them promptly and effectively
- GS6.** complete the assigned tasks timely

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with co-workers</i>	15	30	-	10
PC1. identify own direct and indirect responsibilities, e.g. duties concerning team members	3	5	-	3
PC2. undertake appropriate activities to promote cooperation and good relationships with co-workers	1	2	-	1
PC3. mentor and assist subordinates in the execution of their work responsibilities	1	2	-	1
PC4. identify possible disruptions to work through coordination with the relevant stakeholders and take appropriate preventive measures	3	4	-	2
PC5. follow the recommended practices to avoid and resolve conflicts at work	1	2	-	1
PC6. interact effectively and cooperate with co-workers	1	1	-	-
PC7. manage workplace conflicts, ensuring no impact on work	1	1	-	-
PC8. identify and take responsibility for own mistakes	1	3	-	1
PC9. follow the organizational policies and procedures	1	3	-	-
PC10. follow the appropriate productivity enhancement measures, e.g. Quality circle, 5s, TPM, Six Sigma, JIT, etc.	1	4	-	1
PC11. exchange feedback with peers and other team members to improve the quality of work	1	3	-	-
<i>Communicate effectively with co-workers</i>	9	10	-	5
PC12. follow the organizational policy for the efficient and timely dissemination of information to the authorized personnel	2	3	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. communicate clearly and politely to ensure effective communication with co-workers	1	2	-	-
PC14. practice active listening to understand the conveyed information and instructions	2	1	-	1
PC15. practice openness in work conversations, giving undivided attention to enhance the quality of workplace communication	2	2	-	1
PC16. maintain the professional standard of communication, e.g. avoiding the use of offensive language at work	2	2	-	1
<i>Practice inclusiveness at work</i>	6	10	-	5
PC17. empathize with Persons with Disabilities (PwD)	2	2	-	2
PC18. practice gender equality at work	2	2	-	2
PC19. maintain respectful behaviour and etiquette at work	1	3	-	1
PC20. listen to and acknowledge the feelings, concerns, opinions, and ideas of others	1	3	-	-
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	MIN/N1811
NOS Name	Work collaboratively and effectively in a team
Sector	Mining
Sub-Sector	Mining Operation
Occupation	Placer Mining
NSQF Level	4
Credits	1
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	14/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
MIN/N3303.Perform inspection, testing, calibration and troubleshooting of measuring and control equipment	30	50	0	20	100	50
MIN/N3304.Carry out preventive, breakdown, scheduled and predictive maintenance of mining equipment	30	50	0	20	100	35
MIN/N1810.Follow the applicable health, safety and environmental guidelines (Placer Mining)	30	50	0	20	100	5
MIN/N1811.Work collaboratively and effectively in a team	30	50	0	20	100	5
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	140	230	-	80	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.